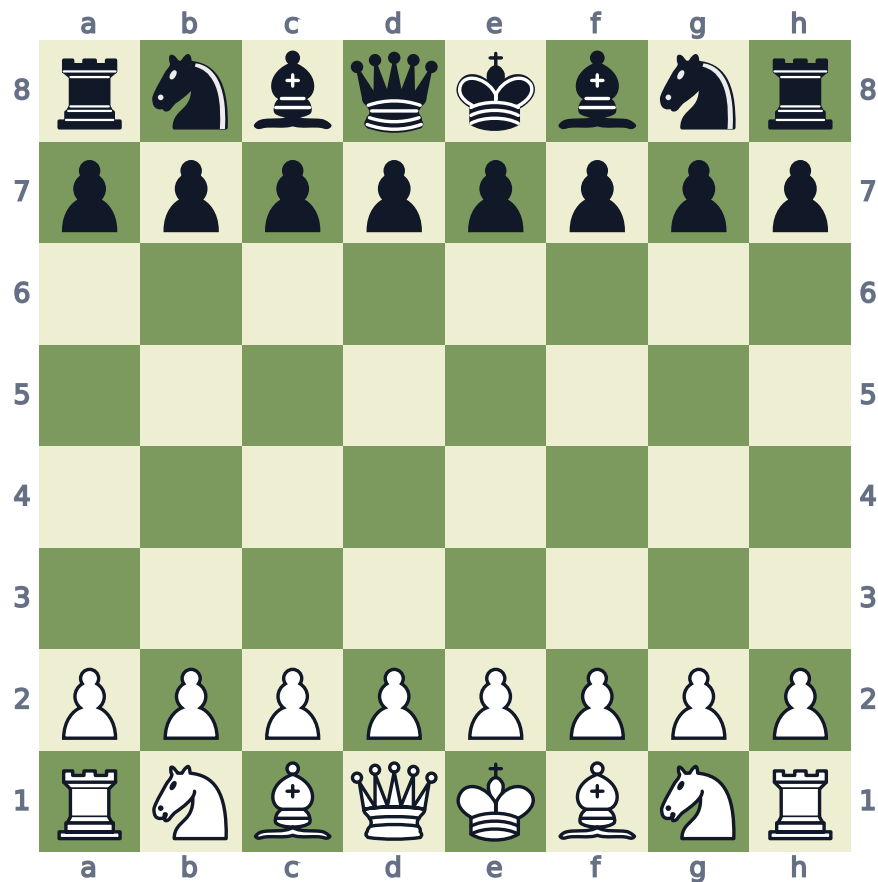


Calculation From the Ground Up

Candidate moves, visualization, pruning, and evaluation



Build a calculation process that is deep enough to be accurate, narrow enough to be practical, and explicit enough to improve after every game.

22 learning units | explanations | board laboratories | review system

ORIENTATION

What this course will change

COURSE PROMISE

Build a calculation process that is deep enough to be accurate, narrow enough to be practical, and explicit enough to improve after every game.

YOU WILL LEARN TO

- Create useful candidates
- Visualize without ghost pieces
- Manage branches and stopping points
- Compare endpoints and verify the final choice
-

HOW TO USE THE COURSE

- Study one learning unit at a time. Say the rule in your own words before continuing.
- At each diagram, hide the model answer. Write candidates and the opponent's best reply.
- Use the rule during a slow game, then record whether it actually changed a decision.
- Review after 1, 3, 7, 14, and 30 days. Retrieval matters more than rereading.
-

SCOPE

This course contains 22 integrated learning units. Principles guide attention, but tactics and exact calculation always decide the move.

CORE METHOD

The decision protocol

USE THIS EVERY TIME

A reliable process is more valuable than a brilliant idea used occasionally. Practice the sequence until it becomes automatic under clock pressure.

1**Build a short candidate list.****2****Order candidates by forcing value and relevance.****3****Visualize one branch at a time.****4****Stop only when forcing play settles.****5****Compare endpoints, reset the board, and verify once.****STOPPING RULE**

Choose when one candidate is sufficiently verified for the clock situation. Before moving, inspect the position once from the opponent's perspective.

COURSE MAP

Learning units

- 01 Visualize the Position After Every Forcing Move
- 02 Branch Where the Opponent Has Real Choices
- 03 Prune Dominated Candidates
- 04 Use Elimination When No Move Looks Best
- 05 Calculate Capture Order Precisely
- 06 Maintain a Material Ledger
- 07 Recognize Transpositions in the Tree
- 08 Calculate Backward From the Desired Result
- 09 Distinguish Forced Moves From Merely Natural Moves
- 10 Use a Verification Pass Before Committing
- 11 Inventory the Position Before Calculating
- 12 Rank Candidates by Reply Restriction
- 13 Calculate the Best Countercheck First
- 14 Audit Every Capture on the Critical Square
- 15 Use Comparison Points in the Tree
- 16 Stop Only at a Quiescent Position
- 17 Return to the Root Between Branches
- 18 Separate Seeing Errors from Judging Errors
- 19 Use Motifs as Search Cues
- 20 Refute Your Own Sacrifice

COURSE MAP

Learning units

- 21 Search for Quiet Tactical Moves
- 22 Treat Move Order as a Tactical Resource

LEARNING UNIT 01

Visualize the Position After Every Forcing Move

THE RULE

Rebuild the board mentally after each move instead of carrying an outdated image forward.

WHY IT WORKS

Calculation errors often come from ghost pieces, forgotten captures, and lines that are no longer blocked. Explicit board updating preserves legal geometry.

RECOGNITION CUES

Several candidates deserve analysis; the position contains forcing continuations; the final evaluation is not obvious

APPLY IT OVER THE BOARD

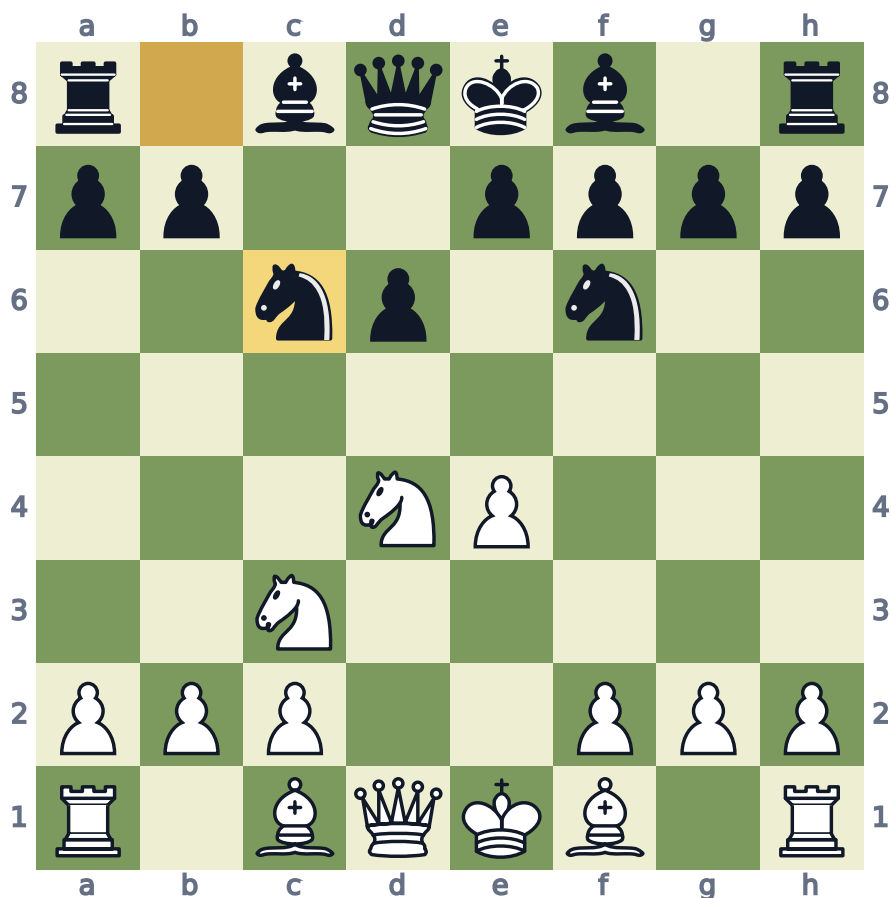
- Build a short candidate list.
- Order candidates by forcing value and relevance.
- Visualize one branch at a time.
- Stop only when forcing play settles.
- Compare endpoints, reset the board, and verify once.
-

BOUNDARY CONDITION

A disciplined method should reduce error, not consume unlimited time. Stop when one move is sufficiently verified for the practical situation.

BOARD LABORATORY 01

Use the idea, then try to disprove it



3. d4 cxd4 4. Nxd4 Nf6 5. Nc3 Nc6

YOUR TASK

Without moving the pieces, apply 'Visualize the Position After Every Forcing Move'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Calculation errors often come from ghost pieces, forgotten captures, and lines that are no longer blocked. Explicit board updating preserves legal geometry. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Analyze ten positions with a written tree using visualize the position after every forcing move. Compare your tree with the game or engine only after finishing independently.

LEARNING UNIT 02

Branch Where the Opponent Has Real Choices

THE RULE

Create calculation branches only at replies that materially change the outcome.

WHY IT WORKS

A tree becomes manageable when forced sequences stay linear and meaningful alternatives create branches. Branching on every legal move overwhelms working memory.

RECOGNITION CUES

Several candidates deserve analysis; the position contains forcing continuations; the final evaluation is not obvious

APPLY IT OVER THE BOARD

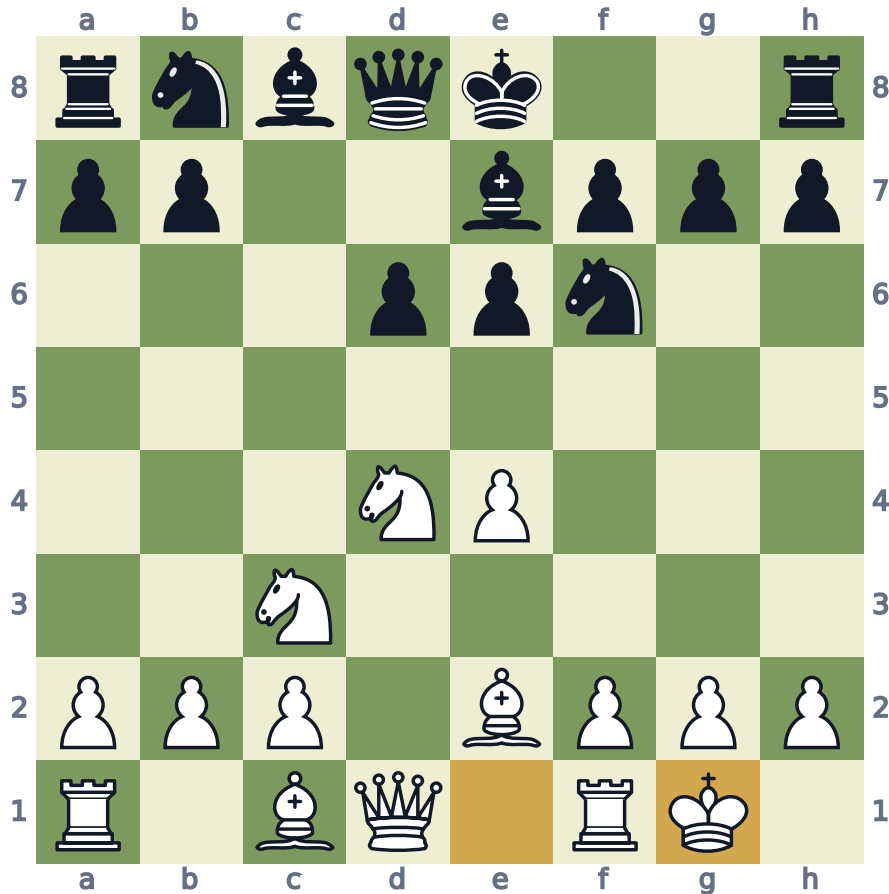
- Build a short candidate list.
- Order candidates by forcing value and relevance.
- Visualize one branch at a time.
- Stop only when forcing play settles.
- Compare endpoints, reset the board, and verify once.
-

BOUNDARY CONDITION

A disciplined method should reduce error, not consume unlimited time. Stop when one move is sufficiently verified for the practical situation.

BOARD LABORATORY 02

Use the idea, then try to disprove it



4... Nf6 5. Nc3 e6 6. Be2 Be7 7. O-O

YOUR TASK

Without moving the pieces, apply 'Branch Where the Opponent Has Real Choices'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: A tree becomes manageable when forced sequences stay linear and meaningful alternatives create branches. Branching on every legal move overwhelms working memory. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Analyze ten positions with a written tree using branch where the opponent has real choices. Compare your tree with the game or engine only after finishing independently.

LEARNING UNIT 03

Prune Dominated Candidates

THE RULE

Stop analyzing a candidate when another move produces the same benefits with fewer concrete risks.

WHY IT WORKS

Calculation time is scarce. If one move is strictly inferior in activity, safety, and tactical soundness, continuing its tree has little decision value.

RECOGNITION CUES

Several candidates deserve analysis; the position contains forcing continuations; the final evaluation is not obvious

APPLY IT OVER THE BOARD

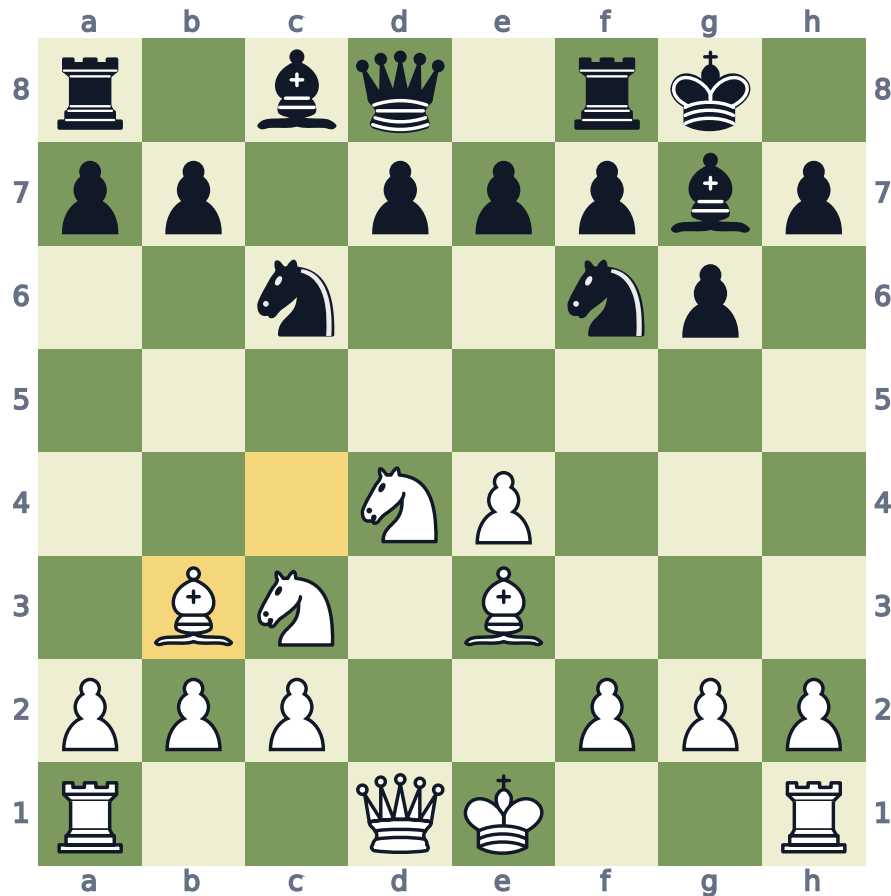
- Build a short candidate list.
- Order candidates by forcing value and relevance.
- Visualize one branch at a time.
- Stop only when forcing play settles.
- Compare endpoints, reset the board, and verify once.
-

BOUNDARY CONDITION

A disciplined method should reduce error, not consume unlimited time. Stop when one move is sufficiently verified for the practical situation.

BOARD LABORATORY 03

Use the idea, then try to disprove it



5... Bg7 6. Be3 Nf6 7. Bc4 O-O 8. Bb3

YOUR TASK

Without moving the pieces, apply 'Prune Dominated Candidates'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Calculation time is scarce. If one move is strictly inferior in activity, safety, and tactical soundness, continuing its tree has little decision value. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Analyze ten positions with a written tree using prune dominated candidates. Compare your tree with the game or engine only after finishing independently.

LEARNING UNIT 04

Use Elimination When No Move Looks Best

THE RULE

Reject candidates with concrete defects until the most resilient option remains.

WHY IT WORKS

Some positions are solved more reliably by disproving moves than proving a single move. Defensive and technical positions especially reward error elimination.

RECOGNITION CUES

Several candidates deserve analysis; the position contains forcing continuations; the final evaluation is not obvious

APPLY IT OVER THE BOARD

- Build a short candidate list.
- Order candidates by forcing value and relevance.
- Visualize one branch at a time.
- Stop only when forcing play settles.
- Compare endpoints, reset the board, and verify once.
-

BOUNDARY CONDITION

A disciplined method should reduce error, not consume unlimited time. Stop when one move is sufficiently verified for the practical situation.

BOARD LABORATORY 04

Use the idea, then try to disprove it



2. Nf3 e6 3. d4 cxd4 4. Nxd4 Nc6

YOUR TASK

Without moving the pieces, apply 'Use Elimination When No Move Looks Best'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Some positions are solved more reliably by disproving moves than proving a single move. Defensive and technical positions especially reward error elimination. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Analyze ten positions with a written tree using use elimination when no move looks best. Compare your tree with the game or engine only after finishing independently.

LEARNING UNIT 05

Calculate Capture Order Precisely

THE RULE

When several exchanges are possible, test different capture and recapture sequences.

WHY IT WORKS

Move order changes which piece occupies the square, which lines open, and whether an intermediate tactic exists. Identical material exchanges can yield different positions.

RECOGNITION CUES

Several candidates deserve analysis; the position contains forcing continuations; the final evaluation is not obvious

APPLY IT OVER THE BOARD

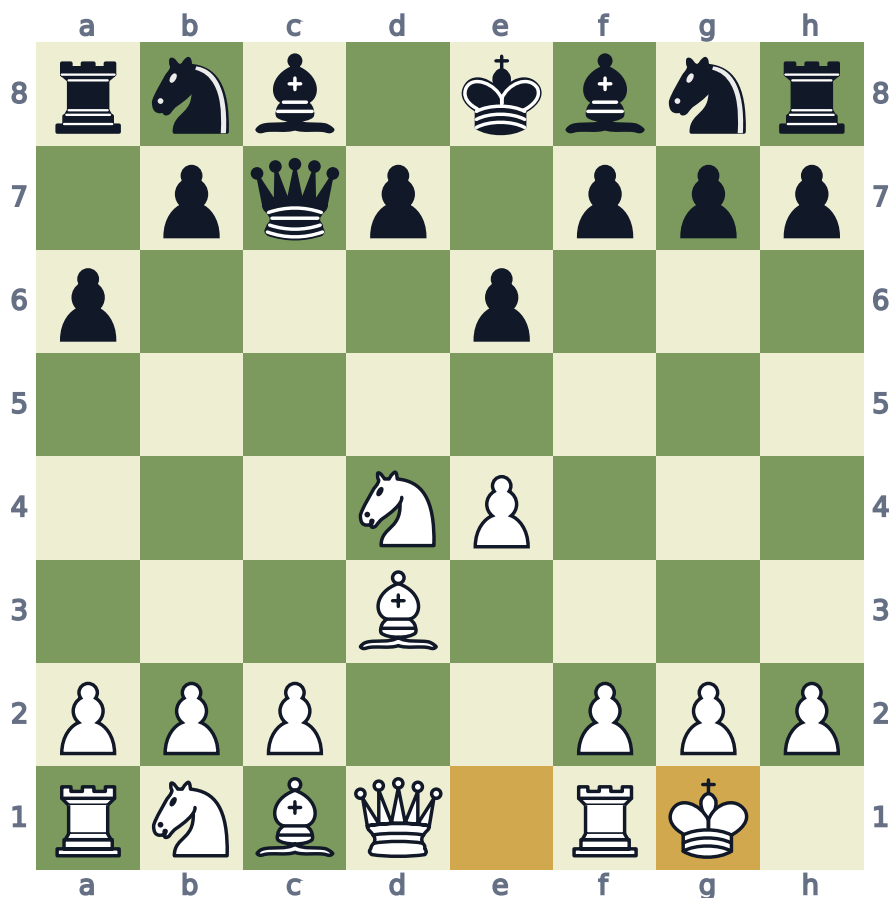
- Build a short candidate list.
- Order candidates by forcing value and relevance.
- Visualize one branch at a time.
- Stop only when forcing play settles.
- Compare endpoints, reset the board, and verify once.
-

BOUNDARY CONDITION

A disciplined method should reduce error, not consume unlimited time. Stop when one move is sufficiently verified for the practical situation.

BOARD LABORATORY 05

Use the idea, then try to disprove it



3... cxd4 4. Nxd4 a6 5. Bd3 Qc7 6. O-O

YOUR TASK

Without moving the pieces, apply 'Calculate Capture Order Precisely'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Move order changes which piece occupies the square, which lines open, and whether an intermediate tactic exists. Identical material exchanges can yield different positions. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Analyze ten positions with a written tree using calculate capture order precisely. Compare your tree with the game or engine only after finishing independently.

LEARNING UNIT 06

Maintain a Material Ledger

THE RULE

Track material changes explicitly during long combinations.

WHY IT WORKS

Memory tends to remember the dramatic sacrifice and forget later recaptures. A simple running ledger prevents false conclusions about the final balance.

RECOGNITION CUES

Several candidates deserve analysis; the position contains forcing continuations; the final evaluation is not obvious

APPLY IT OVER THE BOARD

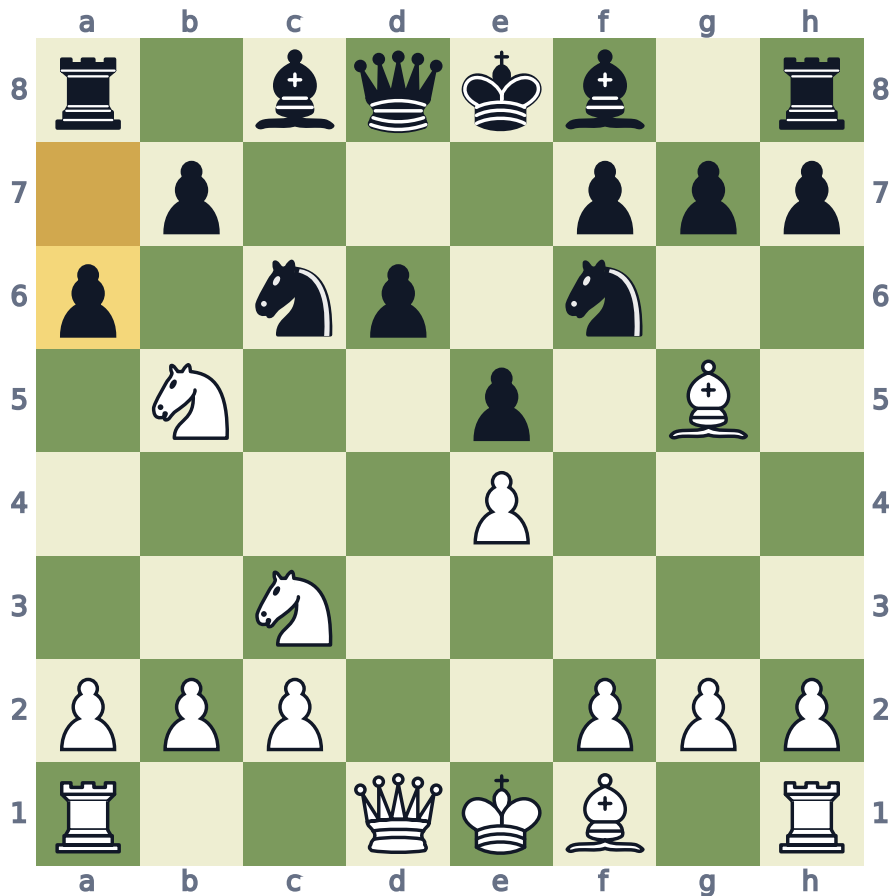
- Build a short candidate list.
- Order candidates by forcing value and relevance.
- Visualize one branch at a time.
- Stop only when forcing play settles.
- Compare endpoints, reset the board, and verify once.
-

BOUNDARY CONDITION

A disciplined method should reduce error, not consume unlimited time. Stop when one move is sufficiently verified for the practical situation.

BOARD LABORATORY 06

Use the idea, then try to disprove it



5. Nc3 e5 6. Ndb5 d6 7. Bg5 a6

YOUR TASK

Without moving the pieces, apply 'Maintain a Material Ledger'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Memory tends to remember the dramatic sacrifice and forget later recaptures. A simple running ledger prevents false conclusions about the final balance. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Analyze ten positions with a written tree using maintain a material ledger. Compare your tree with the game or engine only after finishing independently.

LEARNING UNIT 07

Recognize Transpositions in the Tree

THE RULE

Notice when different move orders reach the same position and evaluate it once.

WHY IT WORKS

Transpositions reduce duplicated work. They also reveal whether a move order allows an extra resource before the common position is reached.

RECOGNITION CUES

Several candidates deserve analysis; the position contains forcing continuations; the final evaluation is not obvious

APPLY IT OVER THE BOARD

- Build a short candidate list.
- Order candidates by forcing value and relevance.
- Visualize one branch at a time.
- Stop only when forcing play settles.
- Compare endpoints, reset the board, and verify once.
-

BOUNDARY CONDITION

A disciplined method should reduce error, not consume unlimited time. Stop when one move is sufficiently verified for the practical situation.

BOARD LABORATORY 07

Use the idea, then try to disprove it



1... e5 2. Nf3 Nc6 3. Bc4 Bc5 4. c3

YOUR TASK

Without moving the pieces, apply 'Recognize Transpositions in the Tree'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Transpositions reduce duplicated work. They also reveal whether a move order allows an extra resource before the common position is reached. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Analyze ten positions with a written tree using recognize transpositions in the tree. Compare your tree with the game or engine only after finishing independently.

LEARNING UNIT 08

Calculate Backward From the Desired Result

THE RULE

When seeking mate, promotion, or a tactical construction, identify the required final geometry and work backward to enabling moves.

WHY IT WORKS

Backward reasoning supplies a target for candidate generation. It clarifies which defender must disappear, which line must open, or which tempo is required.

RECOGNITION CUES

Several candidates deserve analysis; the position contains forcing continuations; the final evaluation is not obvious

APPLY IT OVER THE BOARD

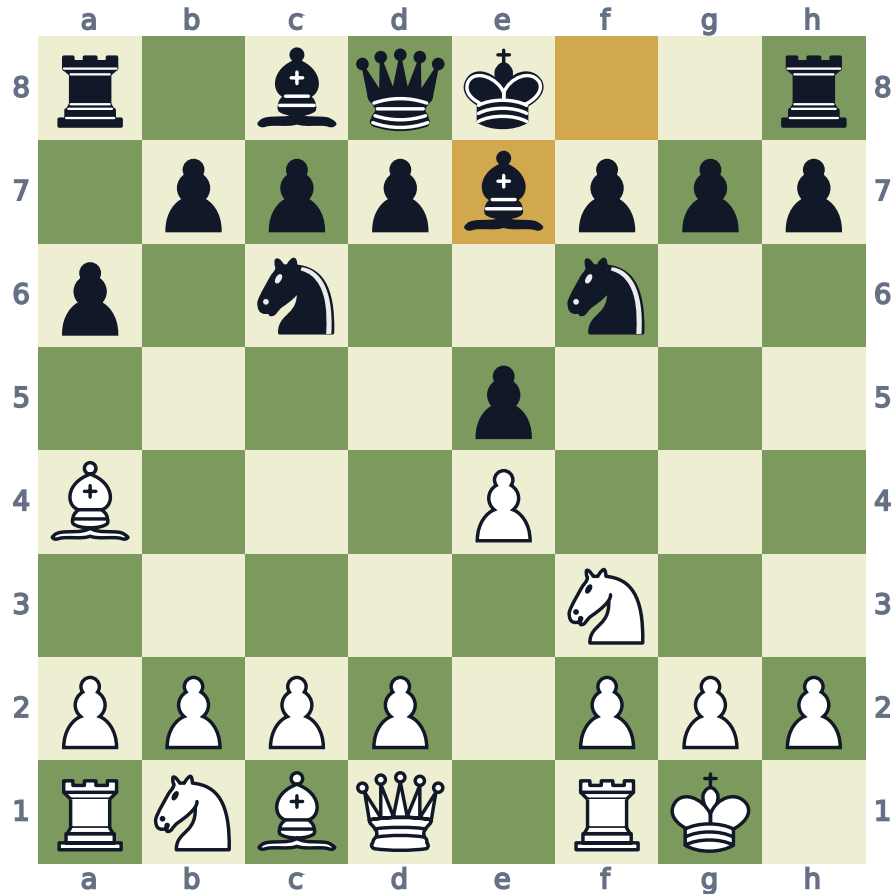
- Build a short candidate list.
- Order candidates by forcing value and relevance.
- Visualize one branch at a time.
- Stop only when forcing play settles.
- Compare endpoints, reset the board, and verify once.
-

BOUNDARY CONDITION

A disciplined method should reduce error, not consume unlimited time. Stop when one move is sufficiently verified for the practical situation.

BOARD LABORATORY 08

Use the idea, then try to disprove it



3. Bb5 a6 4. Ba4 Nf6 5. O-O Be7

YOUR TASK

Without moving the pieces, apply 'Calculate Backward From the Desired Result'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Backward reasoning supplies a target for candidate generation. It clarifies which defender must disappear, which line must open, or which tempo is required. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Analyze ten positions with a written tree using calculate backward from the desired result. Compare your tree with the game or engine only after finishing independently.

LEARNING UNIT 09

Distinguish Forced Moves From Merely Natural Moves

THE RULE

Label replies as forced only when all alternatives fail concretely.

WHY IT WORKS

Calling a natural reply forced creates imaginary combinations. Strong defenders often choose an ugly move, return material, or decline the expected capture.

RECOGNITION CUES

Several candidates deserve analysis; the position contains forcing continuations; the final evaluation is not obvious

APPLY IT OVER THE BOARD

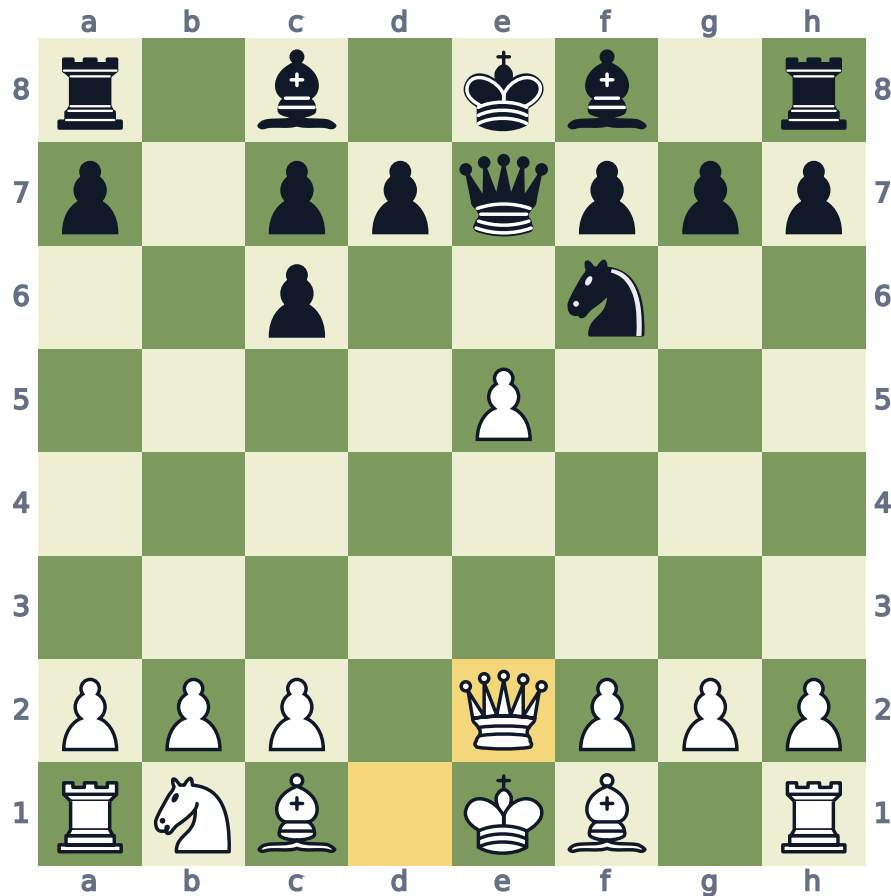
- Build a short candidate list.
- Order candidates by forcing value and relevance.
- Visualize one branch at a time.
- Stop only when forcing play settles.
- Compare endpoints, reset the board, and verify once.
-

BOUNDARY CONDITION

A disciplined method should reduce error, not consume unlimited time. Stop when one move is sufficiently verified for the practical situation.

BOARD LABORATORY 09

Use the idea, then try to disprove it



4... Nf6 5. Nxc6 bxc6 6. e5 Qe7 7. Qe2

YOUR TASK

Without moving the pieces, apply 'Distinguish Forced Moves From Merely Natural Moves'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Calling a natural reply forced creates imaginary combinations. Strong defenders often choose an ugly move, return material, or decline the expected capture. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Analyze ten positions with a written tree using distinguish forced moves from merely natural moves. Compare your tree with the game or engine only after finishing independently.

LEARNING UNIT 10

Use a Verification Pass Before Committing

THE RULE

After selecting a move, inspect the line once from the opponent's perspective.

WHY IT WORKS

A separate verification pass interrupts confirmation bias and catches omitted checks, loose pieces, illegal assumptions, and defensive resources.

RECOGNITION CUES

Several candidates deserve analysis; the position contains forcing continuations; the final evaluation is not obvious

APPLY IT OVER THE BOARD

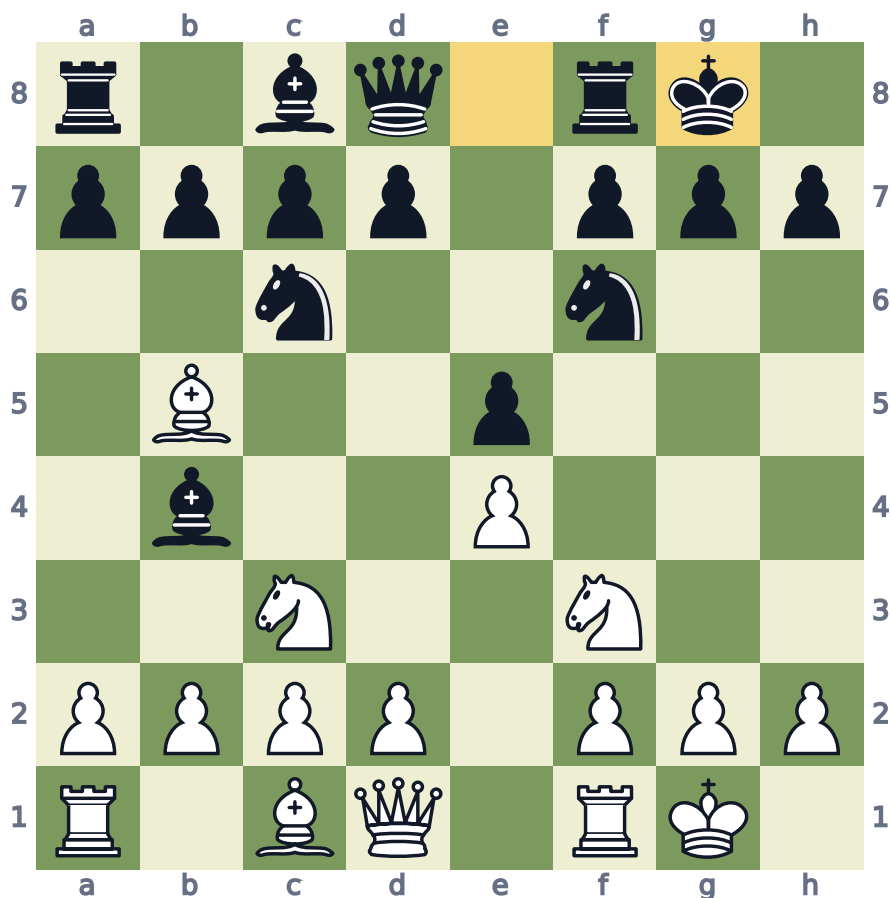
- Build a short candidate list.
- Order candidates by forcing value and relevance.
- Visualize one branch at a time.
- Stop only when forcing play settles.
- Compare endpoints, reset the board, and verify once.
-

BOUNDARY CONDITION

A disciplined method should reduce error, not consume unlimited time. Stop when one move is sufficiently verified for the practical situation.

BOARD LABORATORY 10

Use the idea, then try to disprove it



3. Nc3 Nf6 4. Bb5 Bb4 5. O-O O-O

YOUR TASK

Without moving the pieces, apply 'Use a Verification Pass Before Committing'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: A separate verification pass interrupts confirmation bias and catches omitted checks, loose pieces, illegal assumptions, and defensive resources. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Analyze ten positions with a written tree using use a verification pass before committing. Compare your tree with the game or engine only after finishing independently.

LEARNING UNIT 11

Inventory the Position Before Calculating

THE RULE

Before analyzing moves, record material, king safety, loose pieces, pawn breaks, and immediate threats.

WHY IT WORKS

Calculation is only as good as its starting model. A ten-second inventory prevents entire branches from being built on a missed check, an undefended piece, or an incorrect material count, and it supplies evaluation criteria for the positions reached.

RECOGNITION CUES

the position has several plausible forcing continuations; one branch depends on a long visualization; the opponent has an active tactical reply

APPLY IT OVER THE BOARD

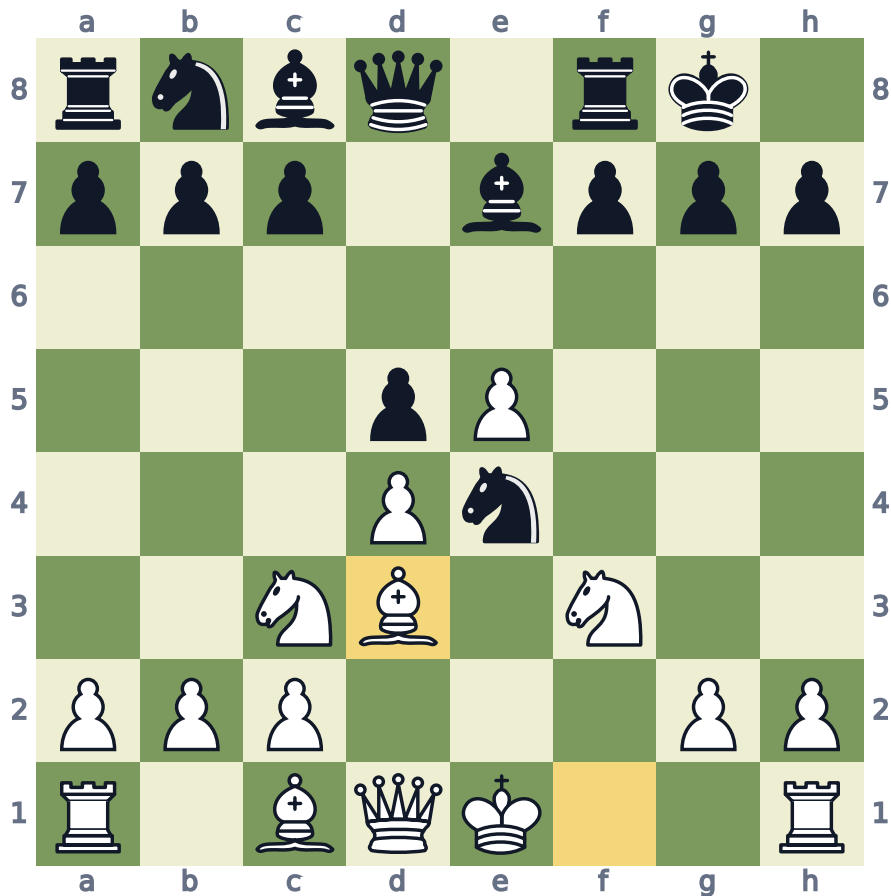
- Build a short candidate list.
- Order candidates by forcing value and relevance.
- Visualize one branch at a time.
- Stop only when forcing play settles.
- Compare endpoints, reset the board, and verify once.
-

BOUNDARY CONDITION

Do not calculate mechanically to a fixed depth. Stop a branch when the position is forced, clearly evaluated, or superseded by a stronger candidate, but never before checking the opponent's forcing replies.

BOARD LABORATORY 11

Use the idea, then try to disprove it



4... Nxe4 5. Nf3 Be7 6. d4 O-O 7. Bd3

YOUR TASK

Without moving the pieces, apply 'Inventory the Position Before Calculating'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Calculation is only as good as its starting model. A ten-second inventory prevents entire branches from being built on a missed check, an undefended piece, or an incorrect material count, and it supplies evaluation criteria for the positions reached. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Inventory the Position Before Calculating'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven...

LEARNING UNIT 12

Rank Candidates by Reply Restriction

THE RULE

Investigate candidates that most restrict the opponent first, while still including sound quiet alternatives.

WHY IT WORKS

A move that allows only one or two serious replies creates a narrower tree and makes calculation more reliable. Restriction is not the same as strength, but it is an efficient ordering device because forcing defenses expose tactical truth quickly.

RECOGNITION CUES

the position has several plausible forcing continuations; one branch depends on a long visualization; the opponent has an active tactical reply

APPLY IT OVER THE BOARD

- Build a short candidate list.
- Order candidates by forcing value and relevance.
- Visualize one branch at a time.
- Stop only when forcing play settles.
- Compare endpoints, reset the board, and verify once.
-

BOUNDARY CONDITION

Do not calculate mechanically to a fixed depth. Stop a branch when the position is forced, clearly evaluated, or superseded by a stronger candidate, but never before checking the opponent's forcing replies.

BOARD LABORATORY 12

Use the idea, then try to disprove it



4. h4 g4 5. Ne5 Nf6 6. Bc4 d5

YOUR TASK

Without moving the pieces, apply 'Rank Candidates by Reply Restriction'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: A move that allows only one or two serious replies creates a narrower tree and makes calculation more reliable. Restriction is not the same as strength, but it is an efficient ordering device because forcing defenses expose tactical truth quickly. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Rank Candidates by Reply Restriction'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven...

LEARNING UNIT 13

Calculate the Best Countercheck First

THE RULE

When attacking, test the opponent's strongest check before assuming they must answer your threat.

WHY IT WORKS

Checks can reverse move order, expose your king, or force a perpetual before your attack lands. Examining the most dangerous countercheck first stress-tests the branch at its weakest point and prevents wishful one-sided calculation.

RECOGNITION CUES

the position has several plausible forcing continuations; one branch depends on a long visualization; the opponent has an active tactical reply

APPLY IT OVER THE BOARD

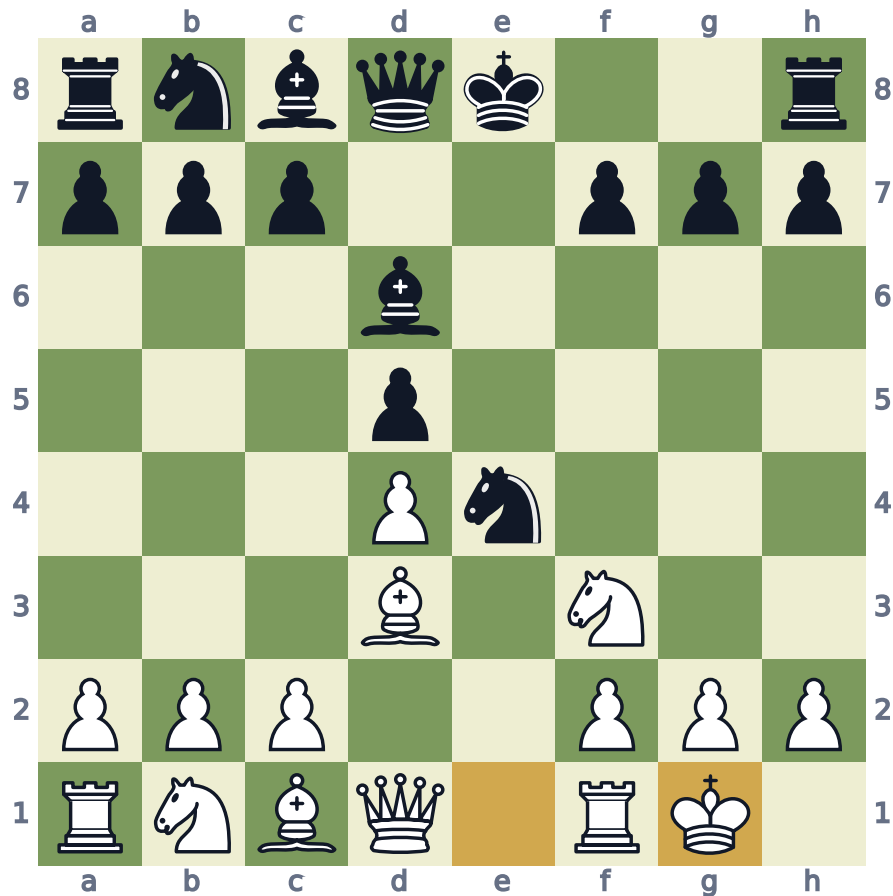
- Build a short candidate list.
- Order candidates by forcing value and relevance.
- Visualize one branch at a time.
- Stop only when forcing play settles.
- Compare endpoints, reset the board, and verify once.
-

BOUNDARY CONDITION

Do not calculate mechanically to a fixed depth. Stop a branch when the position is forced, clearly evaluated, or superseded by a stronger candidate, but never before checking the opponent's forcing replies.

BOARD LABORATORY 13

Use the idea, then try to disprove it



4... Nxe4 5. d4 d5 6. Bd3 Bd6 7. O-O

YOUR TASK

Without moving the pieces, apply 'Calculate the Best Countercheck First'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Checks can reverse move order, expose your king, or force a perpetual before your attack lands. Examining the most dangerous countercheck first stress-tests the branch at its weakest point and prevents wishful one-sided calculation. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Calculate the Best Countercheck First'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days.

LEARNING UNIT 14

Audit Every Capture on the Critical Square

THE RULE

When a line turns on one square, list every piece that can capture there and in what order.

WHY IT WORKS

Exchanges change defenders, open lines, and create intermediate moves. An explicit capture audit prevents the common error of counting attackers only once even though each recapture changes which pieces and pins remain.

RECOGNITION CUES

the position has several plausible forcing continuations; one branch depends on a long visualization; the opponent has an active tactical reply

APPLY IT OVER THE BOARD

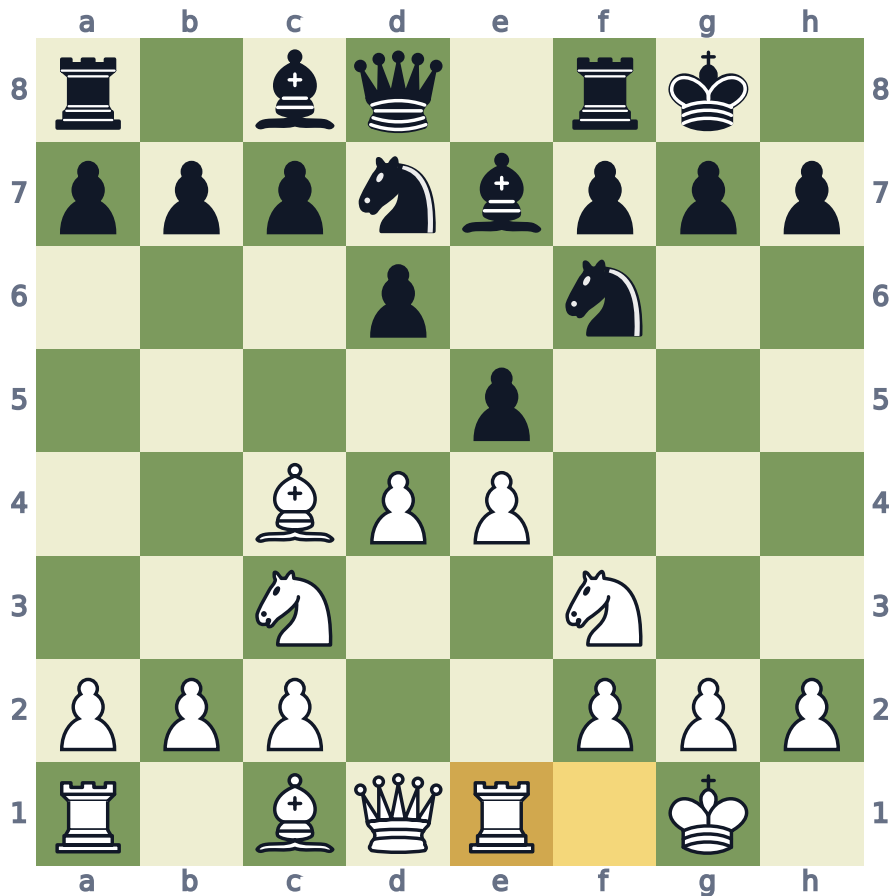
- Build a short candidate list.
- Order candidates by forcing value and relevance.
- Visualize one branch at a time.
- Stop only when forcing play settles.
- Compare endpoints, reset the board, and verify once.
-

BOUNDARY CONDITION

Do not calculate mechanically to a fixed depth. Stop a branch when the position is forced, clearly evaluated, or superseded by a stronger candidate, but never before checking the opponent's forcing replies.

BOARD LABORATORY 14

Use the idea, then try to disprove it



4... Nbd7 5. Bc4 Be7 6. O-O O-O 7. Re1

YOUR TASK

Without moving the pieces, apply 'Audit Every Capture on the Critical Square'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Exchanges change defenders, open lines, and create intermediate moves. An explicit capture audit prevents the common error of counting attackers only once even though each recapture changes which pieces and pins remain. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Audit Every Capture on the Critical Square'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven...

LEARNING UNIT 15

Use Comparison Points in the Tree

THE RULE

Compare candidate branches at stable positions using the same evaluation checklist.

WHY IT WORKS

Human memory distorts branches when they are judged at different depths or by different criteria. Stopping at comparable positions and checking material, kings, activity, structure, and initiative makes the choice evidence-based.

RECOGNITION CUES

the position has several plausible forcing continuations; one branch depends on a long visualization; the opponent has an active tactical reply

APPLY IT OVER THE BOARD

- Build a short candidate list.
- Order candidates by forcing value and relevance.
- Visualize one branch at a time.
- Stop only when forcing play settles.
- Compare endpoints, reset the board, and verify once.
-

BOUNDARY CONDITION

Do not calculate mechanically to a fixed depth. Stop a branch when the position is forced, clearly evaluated, or superseded by a stronger candidate, but never before checking the opponent's forcing replies.

BOARD LABORATORY 15

Use the idea, then try to disprove it



1... d5 2. exd5 Qxd5 3. Nc3 Qd8 4. d4

YOUR TASK

Without moving the pieces, apply 'Use Comparison Points in the Tree'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Human memory distorts branches when they are judged at different depths or by different criteria. Stopping at comparable positions and checking material, kings, activity, structure, and initiative makes the choice evidence-based. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Use Comparison Points in the Tree'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days; scor...

LEARNING UNIT 16

Stop Only at a Quiescent Position

THE RULE

Do not evaluate a branch while an immediate forcing sequence is still unresolved.

WHY IT WORKS

A temporary material gain may disappear to a check, recapture, or zwischenzug. Extending analysis until checks, major captures, and direct threats settle produces an evaluation of a real position rather than a tactical snapshot.

RECOGNITION CUES

the position has several plausible forcing continuations; one branch depends on a long visualization; the opponent has an active tactical reply

APPLY IT OVER THE BOARD

- Build a short candidate list.
- Order candidates by forcing value and relevance.
- Visualize one branch at a time.
- Stop only when forcing play settles.
- Compare endpoints, reset the board, and verify once.
-

BOUNDARY CONDITION

Do not calculate mechanically to a fixed depth. Stop a branch when the position is forced, clearly evaluated, or superseded by a stronger candidate, but never before checking the opponent's forcing replies.

BOARD LABORATORY 16

Use the idea, then try to disprove it



3. Nc3 Nf6 4. e5 Nfd7 5. f4 c5

YOUR TASK

Without moving the pieces, apply 'Stop Only at a Quiescent Position'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: A temporary material gain may disappear to a check, recapture, or zwischenzug. Extending analysis until checks, major captures, and direct threats settle produces an evaluation of a real position rather than a tactical snapshot. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Stop Only at a Quiescent Position'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days; scor...

LEARNING UNIT 17

Return to the Root Between Branches

THE RULE

After calculating one candidate, mentally reset every piece before starting the next.

WHY IT WORKS

Visualization residue causes pieces from one imagined line to leak into another. Reconstructing the original position restores accurate geometry and makes comparisons between candidates independent.

RECOGNITION CUES

the position has several plausible forcing continuations; one branch depends on a long visualization; the opponent has an active tactical reply

APPLY IT OVER THE BOARD

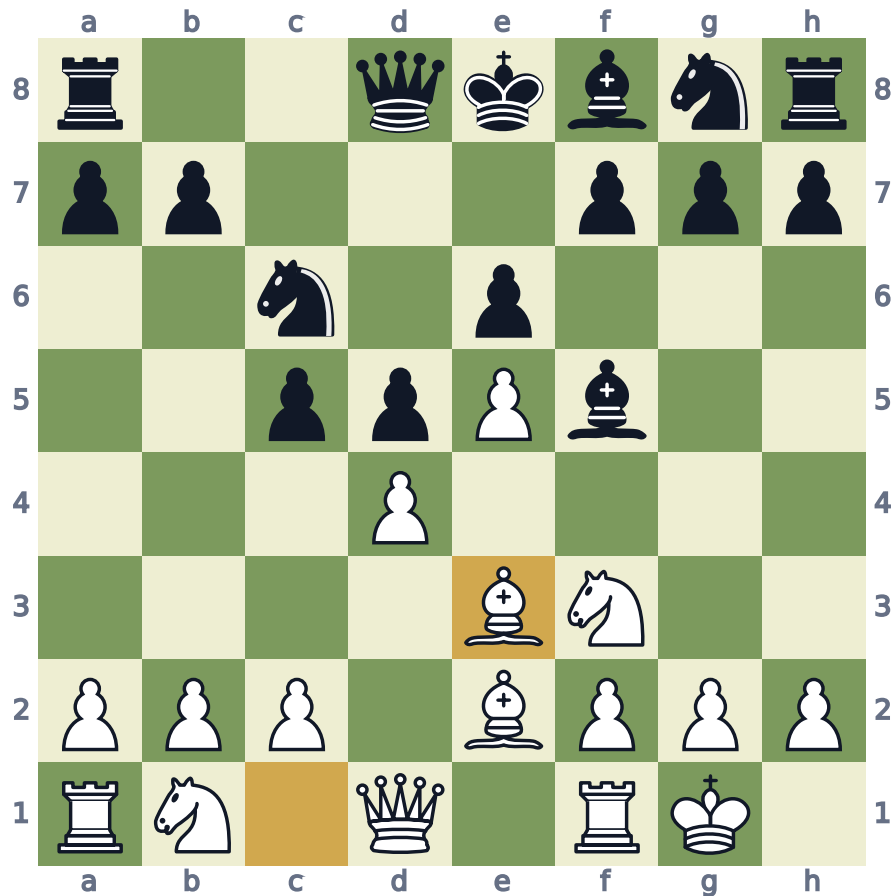
- Build a short candidate list.
- Order candidates by forcing value and relevance.
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BOUNDARY CONDITION

Do not calculate mechanically to a fixed depth. Stop a branch when the position is forced, clearly evaluated, or superseded by a stronger candidate, but never before checking the opponent's forcing replies.

BOARD LABORATORY 17

Use the idea, then try to disprove it



4... e6 5. Be2 c5 6. O-O Nc6 7. Be3

YOUR TASK

Without moving the pieces, apply 'Return to the Root Between Branches'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Visualization residue causes pieces from one imagined line to leak into another. Reconstructing the original position restores accurate geometry and makes comparisons between candidates independent. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Return to the Root Between Branches'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days.

LEARNING UNIT 18

Separate Seeing Errors from Judging Errors

THE RULE

Classify a calculation miss as visualization, candidate generation, or evaluation-not simply 'a blunder.'

WHY IT WORKS

Different failures require different training. Misplacing a piece calls for visualization drills; never considering the move calls for candidate work; reaching the position but judging it wrongly calls for evaluation study.

RECOGNITION CUES

the position has several plausible forcing continuations; one branch depends on a long visualization; the opponent has an active tactical reply

APPLY IT OVER THE BOARD

- Build a short candidate list.
- Order candidates by forcing value and relevance.
- Visualize one branch at a time.
- Stop only when forcing play settles.
- Compare endpoints, reset the board, and verify once.
-

BOUNDARY CONDITION

Do not calculate mechanically to a fixed depth. Stop a branch when the position is forced, clearly evaluated, or superseded by a stronger candidate, but never before checking the opponent's forcing replies.

BOARD LABORATORY 18

Use the idea, then try to disprove it



1... c5 2. Nf3 d6 3. d4 cxd4 4. Nxd4

YOUR TASK

Without moving the pieces, apply 'Separate Seeing Errors from Judging Errors'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Different failures require different training. Misplacing a piece calls for visualization drills; never considering the move calls for candidate work; reaching the position but judging it wrongly calls for evaluation study. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Separate Seeing Errors from Judging Errors'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days.

LEARNING UNIT 19

Use Motifs as Search Cues

THE RULE

Let tactical patterns tell you where to calculate, never what the final answer must be.

WHY IT WORKS

Pattern recognition compresses the search by highlighting pins, alignments, and loose pieces. Treating the pattern as a hypothesis preserves the speed advantage while calculation checks whether the exact position supports it.

RECOGNITION CUES

the position has several plausible forcing continuations; one branch depends on a long visualization; the opponent has an active tactical reply

APPLY IT OVER THE BOARD

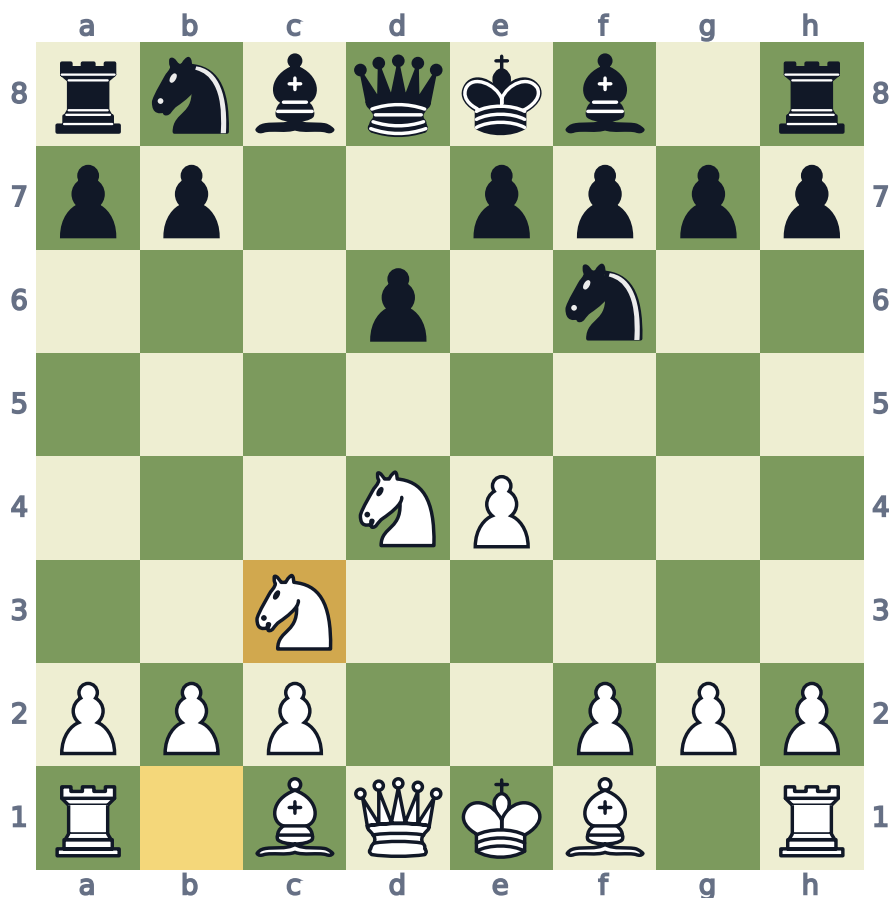
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BOUNDARY CONDITION

Do not calculate mechanically to a fixed depth. Stop a branch when the position is forced, clearly evaluated, or superseded by a stronger candidate, but never before checking the opponent's forcing replies.

BOARD LABORATORY 19

Use the idea, then try to disprove it



2... d6 3. d4 cxd4 4. Nxd4 Nf6 5. Nc3

YOUR TASK

Without moving the pieces, apply 'Use Motifs as Search Cues'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Pattern recognition compresses the search by highlighting pins, alignments, and loose pieces. Treating the pattern as a hypothesis preserves the speed advantage while calculation checks whether the exact position supports it. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Use Motifs as Search Cues'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days; scor...

LEARNING UNIT 20

Refute Your Own Sacrifice

THE RULE

Before sacrificing, actively search for the defender's least cooperative response.

WHY IT WORKS

Attackers naturally calculate attractive replies that confirm the idea. Looking for declined sacrifices, queen trades, counterchecks, and cold defensive retreats counters that bias and reveals whether the compensation is forced or merely hoped for.

RECOGNITION CUES

the position has several plausible forcing continuations; one branch depends on a long visualization; the opponent has an active tactical reply

APPLY IT OVER THE BOARD

- Build a short candidate list.
- Order candidates by forcing value and relevance.
- Visualize one branch at a time.
- Stop only when forcing play settles.
- Compare endpoints, reset the board, and verify once.
-

BOUNDARY CONDITION

Do not calculate mechanically to a fixed depth. Stop a branch when the position is forced, clearly evaluated, or superseded by a stronger candidate, but never before checking the opponent's forcing replies.

BOARD LABORATORY 20

Use the idea, then try to disprove it



4. Nxd4 Nf6 5. Nc3 g6 6. Be3 Bg7

YOUR TASK

Without moving the pieces, apply 'Refute Your Own Sacrifice'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Attackers naturally calculate attractive replies that confirm the idea. Looking for declined sacrifices, queen trades, counterchecks, and cold defensive retreats counters that bias and reveals whether the compensation is forced or merely hoped for. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Refute Your Own Sacrifice'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days; scor...

LEARNING UNIT 21

Search for Quiet Tactical Moves

THE RULE

After checking forcing moves, look for a quiet move that creates two threats or removes the only defense.

WHY IT WORKS

Some combinations are forced by restriction rather than check. A quiet move can dominate escape squares, unpin an attacker, or add a second threat, leaving the defender without a move that answers everything.

RECOGNITION CUES

the position has several plausible forcing continuations; one branch depends on a long visualization; the opponent has an active tactical reply

APPLY IT OVER THE BOARD

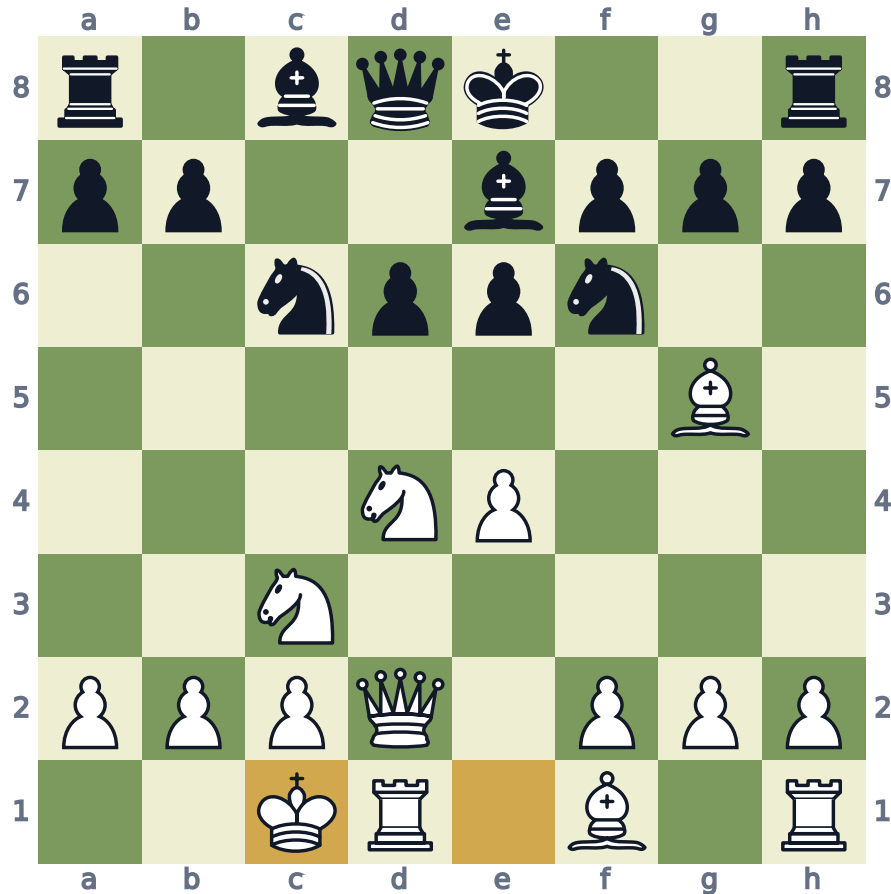
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- Order candidates by forcing value and relevance.
- Visualize one branch at a time.
- Stop only when forcing play settles.
- Compare endpoints, reset the board, and verify once.
-

BOUNDARY CONDITION

Do not calculate mechanically to a fixed depth. Stop a branch when the position is forced, clearly evaluated, or superseded by a stronger candidate, but never before checking the opponent's forcing replies.

BOARD LABORATORY 21

Use the idea, then try to disprove it



5... Nc6 6. Bg5 e6 7. Qd2 Be7 8. O-O-O

YOUR TASK

Without moving the pieces, apply 'Search for Quiet Tactical Moves'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: Some combinations are forced by restriction rather than check. A quiet move can dominate escape squares, unpin an attacker, or add a second threat, leaving the defender without a move that answers everything. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Search for Quiet Tactical Moves'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days; scor...

LEARNING UNIT 22

Treat Move Order as a Tactical Resource

THE RULE

When several useful actions exist, play the one that gains a tempo or denies a defensive setup first.

WHY IT WORKS

The same final arrangement can be winning or harmless depending on sequence. Intermediate checks, attacks on the queen, and threats force replies while you complete your plan, effectively making one move serve two purposes.

RECOGNITION CUES

the position has several plausible forcing continuations; one branch depends on a long visualization; the opponent has an active tactical reply

APPLY IT OVER THE BOARD

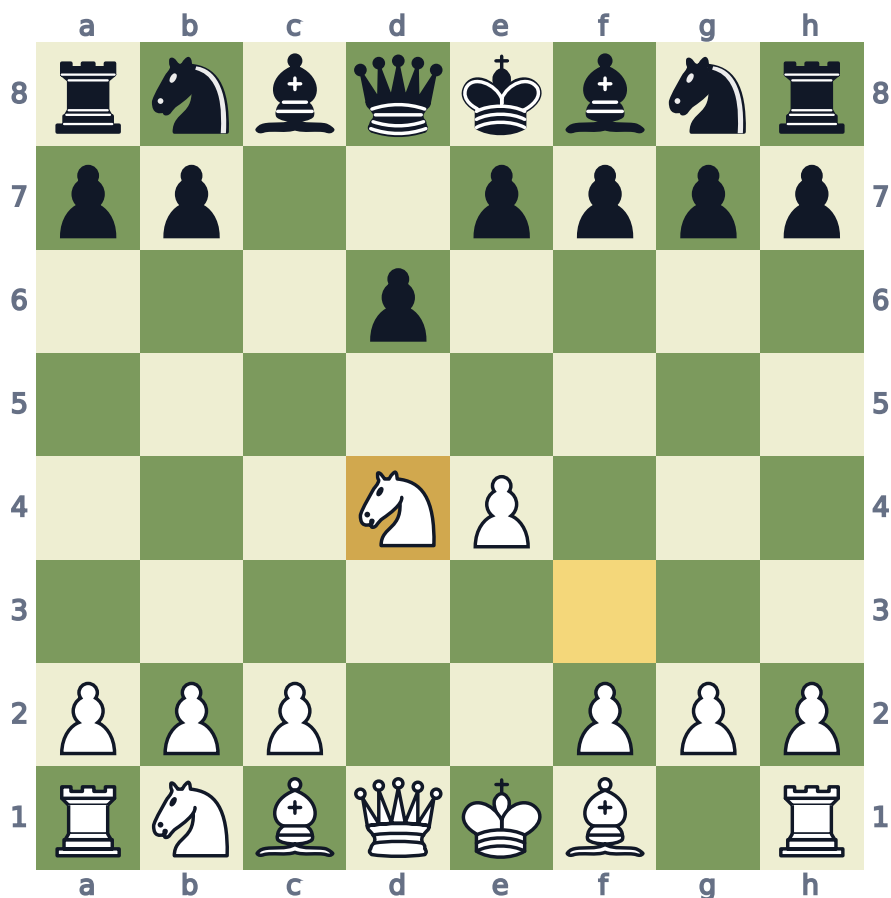
- Build a short candidate list.
- Order candidates by forcing value and relevance.
- Visualize one branch at a time.
- Stop only when forcing play settles.
- Compare endpoints, reset the board, and verify once.
-

BOUNDARY CONDITION

Do not calculate mechanically to a fixed depth. Stop a branch when the position is forced, clearly evaluated, or superseded by a stronger candidate, but never before checking the opponent's forcing replies.

BOARD LABORATORY 22

Use the idea, then try to disprove it



1... c5 2. Nf3 d6 3. d4 cxd4 4. Nxd4

YOUR TASK

Without moving the pieces, apply 'Treat Move Order as a Tactical Resource'. List two or three candidates, the opponent's strongest forcing reply, and the feature that makes the principle relevant or irrelevant here.

MODEL ANSWER STANDARD

A strong answer does more than name the rule. It explains the mechanism: The same final arrangement can be winning or harmless depending on sequence. Intermediate checks, attacks on the queen, and threats force replies while you complete your plan, effectively making one move serve two purposes. Then it verifies the best opposing reply and evaluates the resulting material, king safety, activity, structure, and initiative.

TRAINING TRANSFER

Collect twelve positions connected to 'Treat Move Order as a Tactical Resource'. For each, write three candidates, the opponent's strongest reply, and a one-sentence evaluation. Revisit the same positions without notes after three and seven days.

RETRIEVAL TEST

Can you explain the course without looking?

1

State the five-step decision protocol from memory.

2

Name three recognition cues that should make you slow down.

3

Choose five learning units and explain the causal mechanism behind each rule.

4

For each chosen rule, give one position where it should not be applied mechanically.

5

Describe the opponent's strongest practical resource in your most recent serious game.

6

Name one recurring error and the exact training drill that addresses its cause.

SCORING

2 points: accurate rule plus mechanism. 1 point: useful but incomplete. 0 points: slogan only. Restudy any item below 2 and retrieve it again tomorrow.

TRAINING PLAN

Four weeks to make the ideas usable

WEEK 1 - UNDERSTAND

Study the first quarter of the course. After every lesson, explain why the rule works and when it fails.

WEEK 2 - RECOGNIZE

Study the second quarter. Mix old and new diagrams so the name of the motif is not given in advance.

WEEK 3 - CALCULATE

Finish the course. Use a clock and write candidates, best defense, and endpoint evaluation.

WEEK 4 - TRANSFER

Play two slow games. Mark every moment when the course's protocol changed your decision, then analyze unaided.

LONG-TERM REVIEW

Return after 7, 14, and 30 days. Do not reread first. Attempt the retrieval test, review only missed items, and find one example from your own games.

REFERENCE

How this course fits the library

COURSE ROLE

Build a calculation process that is deep enough to be accurate, narrow enough to be practical, and explicit enough to improve after every game.

LIBRARY PRINCIPLE

Strong chess comes from a loop: notice the right features, generate serious candidates, calculate the opponent's best resistance, evaluate the destination position, and learn from the decision process.

ABOUT THE DIAGRAMS

Boards use Cburnett-style vector chess pieces with solid dark pieces and high-contrast olive and cream squares. Piece artwork is credited to Colin M. L. Burnett and the Lichess Cburnett set under GPLv2+.

EDITION NOTES

First edition, September 2026. 22 learning units. Explanations and exercises are designed for personal study and should be tested against actual games and concrete calculation.

THE FINAL QUESTION

What will you do differently on your very next move? If the course has not changed a behavior, choose one rule, attach it to a visible board cue, and practice it in five slow games.